

$$y = \cos^4(x)$$

$$\Rightarrow y' = 4(\cos^3(x) \cdot (-\sin(x)))$$

$$y' = -4(\cos^3(x)(\sin(x)))$$

$$x = \sin(\sqrt{x^3+1})$$

$$\Rightarrow x' = \cos(\sqrt{x^3+1}) \left(\frac{2x}{2\sqrt{x^3+1}} \right) ~~\left(\frac{2x}{2\sqrt{x^3+1}} \right)~~$$

Simplify:

$$x' = \cos(\sqrt{x^3+1}) \left(\frac{x}{\sqrt{x^3+1}} \right) ~~\left(\frac{x}{\sqrt{x^3+1}} \right)~~$$

$$x' = \frac{x \cdot \cos(\sqrt{x^3+1})}{\sqrt{x^3+1}}$$

